

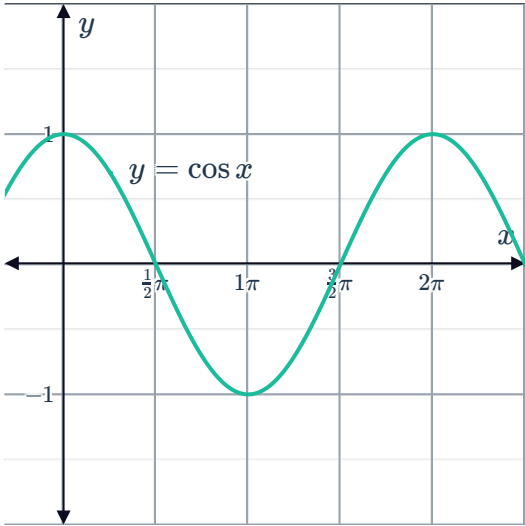
The sine and cosine functions and vertical dilations

1

a

θ	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos \theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e [-4, 4]

2

a $(-\infty, \infty)$

b [-1, 1]

3

a $(-\infty, \infty)$

b [-2, 2]

4 8

5

a $y = 2 \sin x$

b $y = -\sin x$

c $y = 3 \cos x$

d $y = -1.5 \cos x$

e $y = 3 \sin x$

f $y = 2.5 \sin x$

6 $\frac{3\pi}{2}$

7

a 8

b 0

c -8

8 5



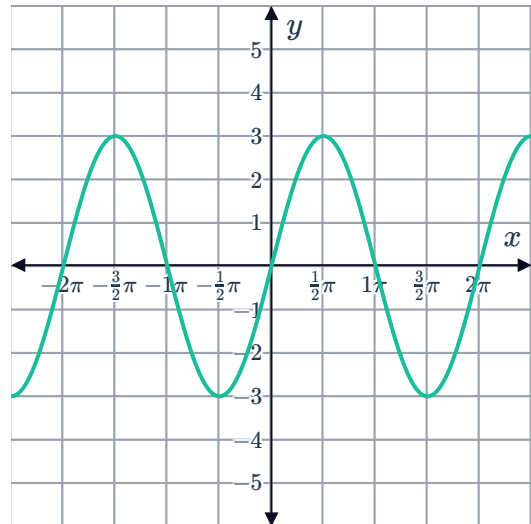
9 $-2 \sin x$

10

a

i $A = 3$

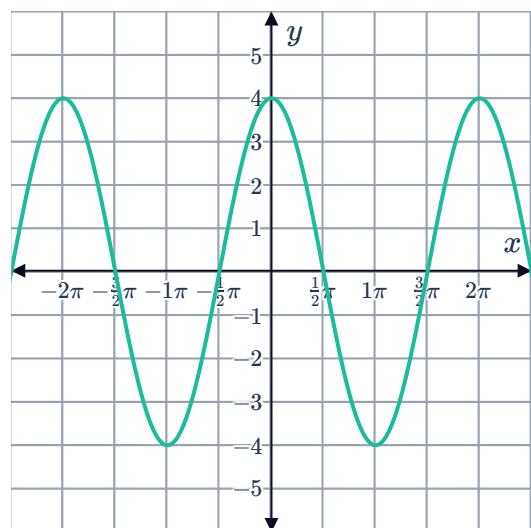
ii



b

i $A = 4$

ii

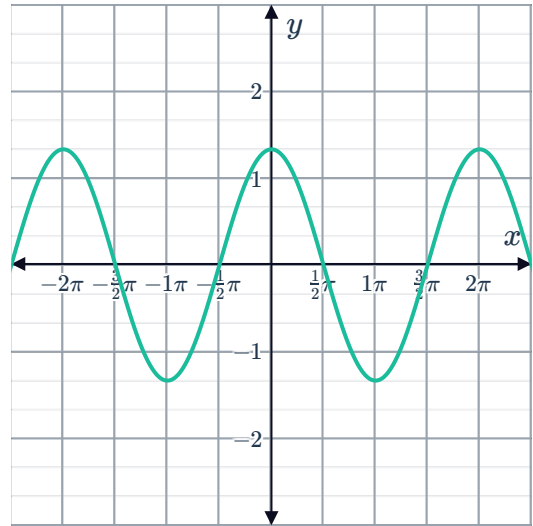


c

i $A = \frac{4}{3}$



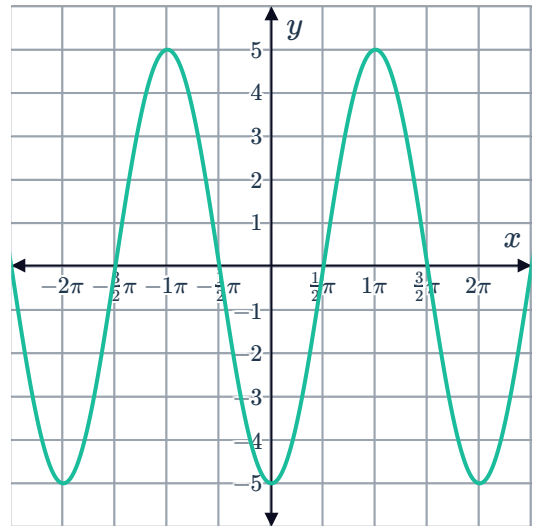
ii



d

i $A = 5$

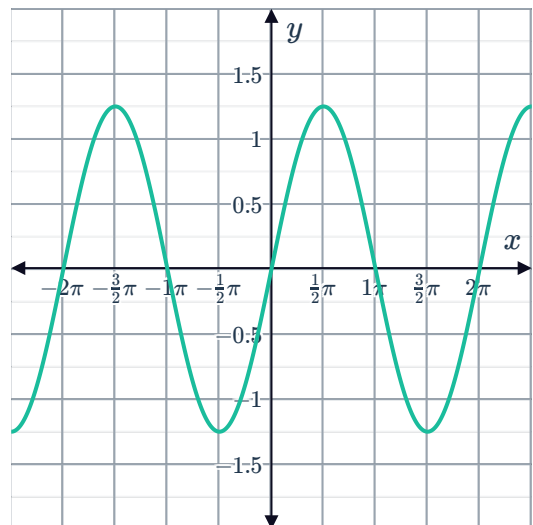
ii



e

i $\frac{5}{4}$

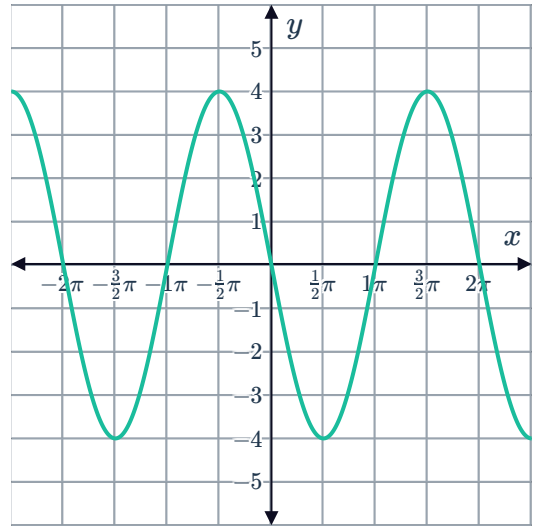
ii



f

i $A = 4$ 

ii



11

a 3

b -3

c 3

d A vertical dilation by a factor of 3 and reflection about the x -axis.

Vertical translations of sine and cosine functions

12 Translated 2 units downwards.

13 Translated 2 units upwards.

14 Translated 4 units upwards

15

a $y = \sin x + 4$ b $y = \sin x - 2$ c $y = \sin x - 4$ d $y = \sin x + 1$

16

a $y = \cos x + 3$ b $y = \cos x - 4$ c $y = \cos x - 3$ d $y = \cos x + 2$ 17 Reflection about the x -axis, then translation 3 units upwards.18 $y = 2$

19

a Translated 4 units upwards.

b 360°



20

a

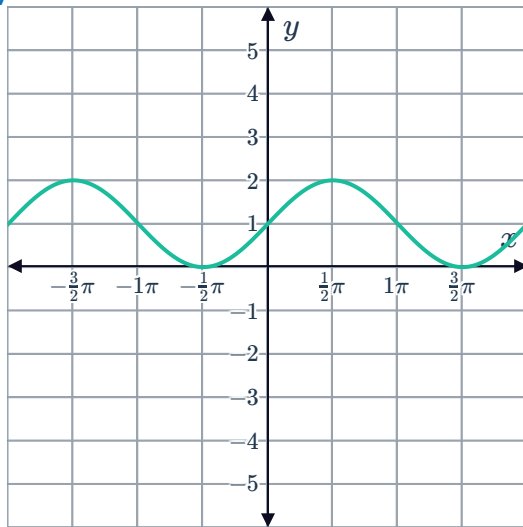
i 2π

ii 1

iii 2

iv 0

v



b

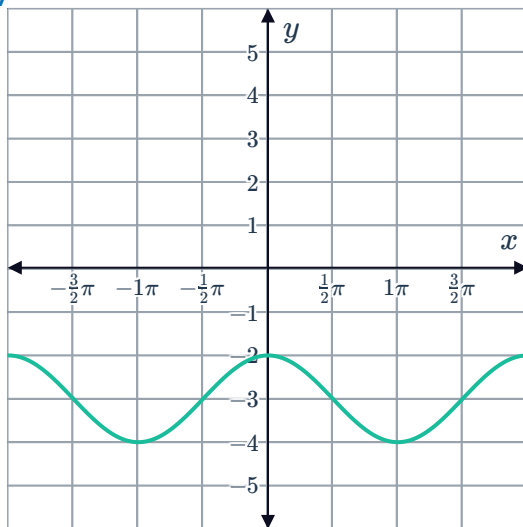
i 2π

ii 1

iii -2

iv -4

v



c

i 2π

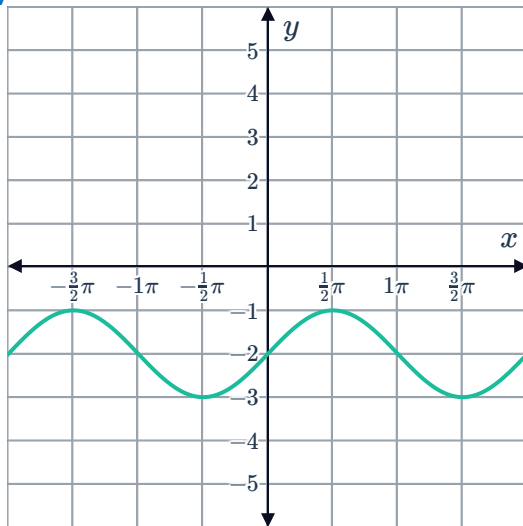
ii 1



iii -1

iv -3

v



d

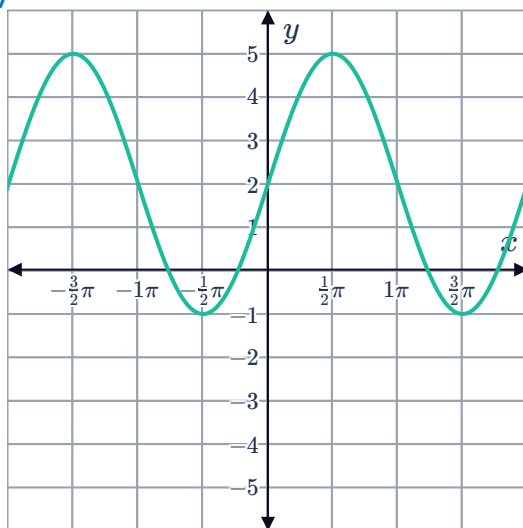
i 2π

ii 3

iii 5

iv -1

v



e

i 2π

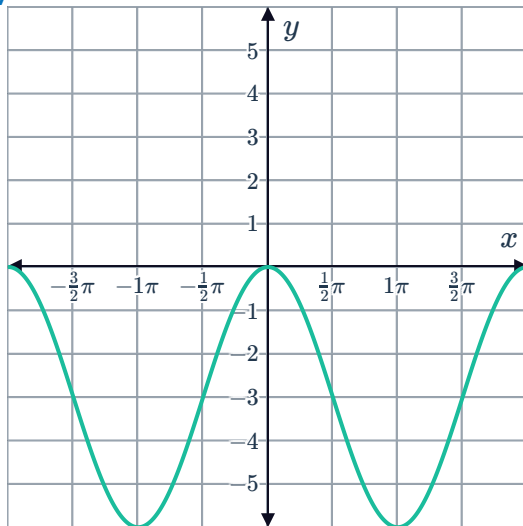
ii 3



iii 0

iv -6

v



f

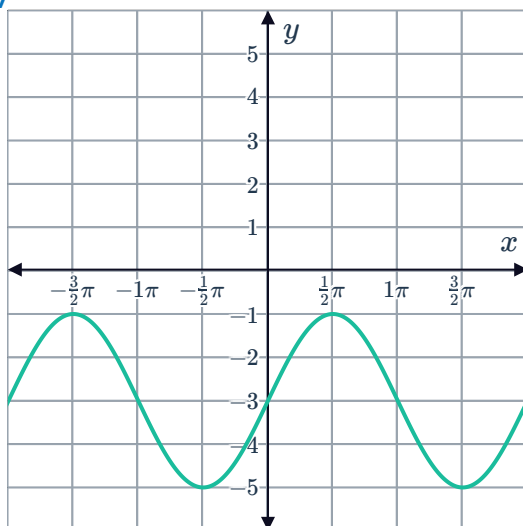
i 2π

ii 2

iii -1

iv -5

v



21



a

i 3

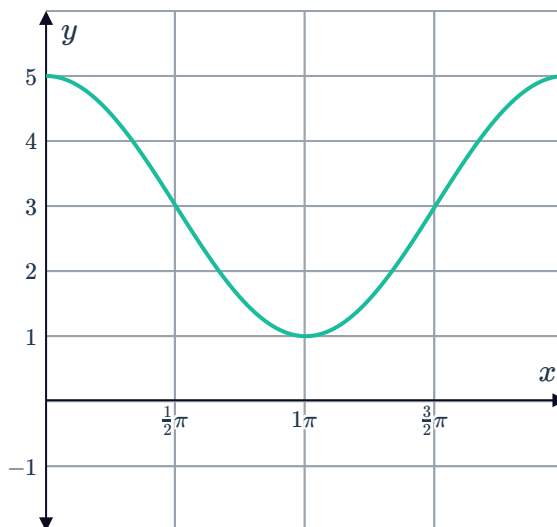
ii 2

iii 5

iv 1

v A vertical dilation by a factor of 2, then translated 3 units upwards.

vi



b

i -3

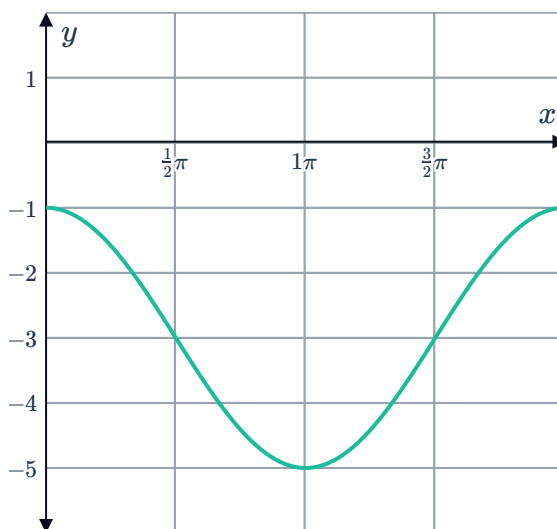
ii -1

iii -5

iv 2

v A vertical dilation by a factor of 2, then translated 3 units downwards.

vi



c

i 2

ii 6

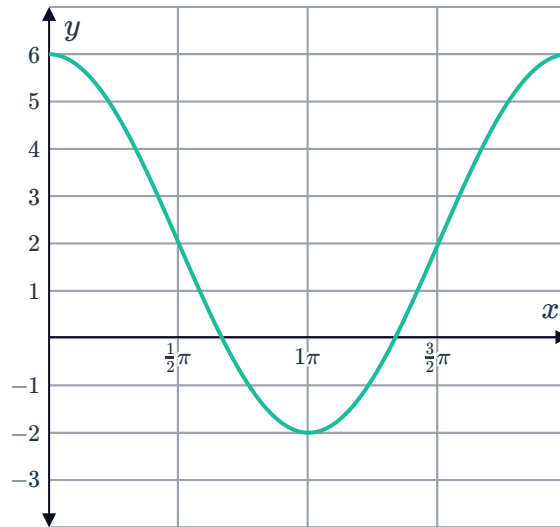


iii -2

iv 4

v A vertical dilation by a factor of 4, a reflection about the x -axis and then translated 2 units upwards.

vi



d

i 2

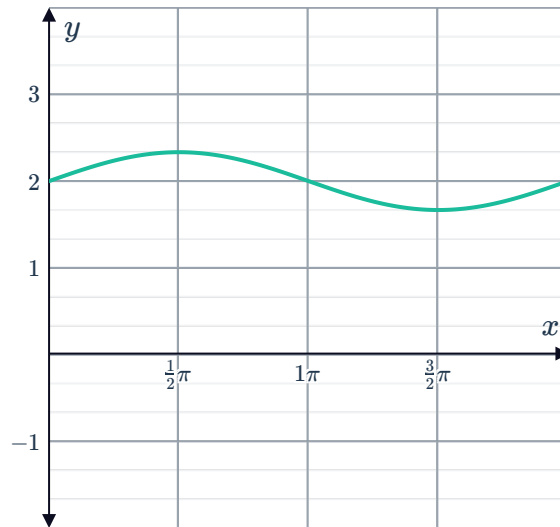
ii $\frac{7}{3}$

iii $\frac{5}{3}$

iv $\frac{1}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



22

a $y = \cos x + 9$

b 10

23

a

i 5

ii -1

b

i 7

ii 1

c

i 2

ii 1

d

i 2

ii $\frac{1}{2}$



24 $2 \sin x + 2$

25 $8 \cos x - 2$

The tangent function and vertical dilations

26

a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k .

b $(-\infty, \infty)$

c $(-\infty, \infty)$

27

a $g(x) = -3 \tan x$

b $f(x) = \frac{1}{5} \tan x$

28

a

i $\frac{1}{4}$

ii Compressed

b

i 6

ii Stretched

c

i $\frac{5}{3}$

ii Stretched

d

i 0.8

ii Compressed

29

a Reflection about the x -axis or reflection about the y -axis.

b $y = -\tan x$

30

a $f(x) = 2 \tan x$

b $f(x) = 0.5 \tan x$

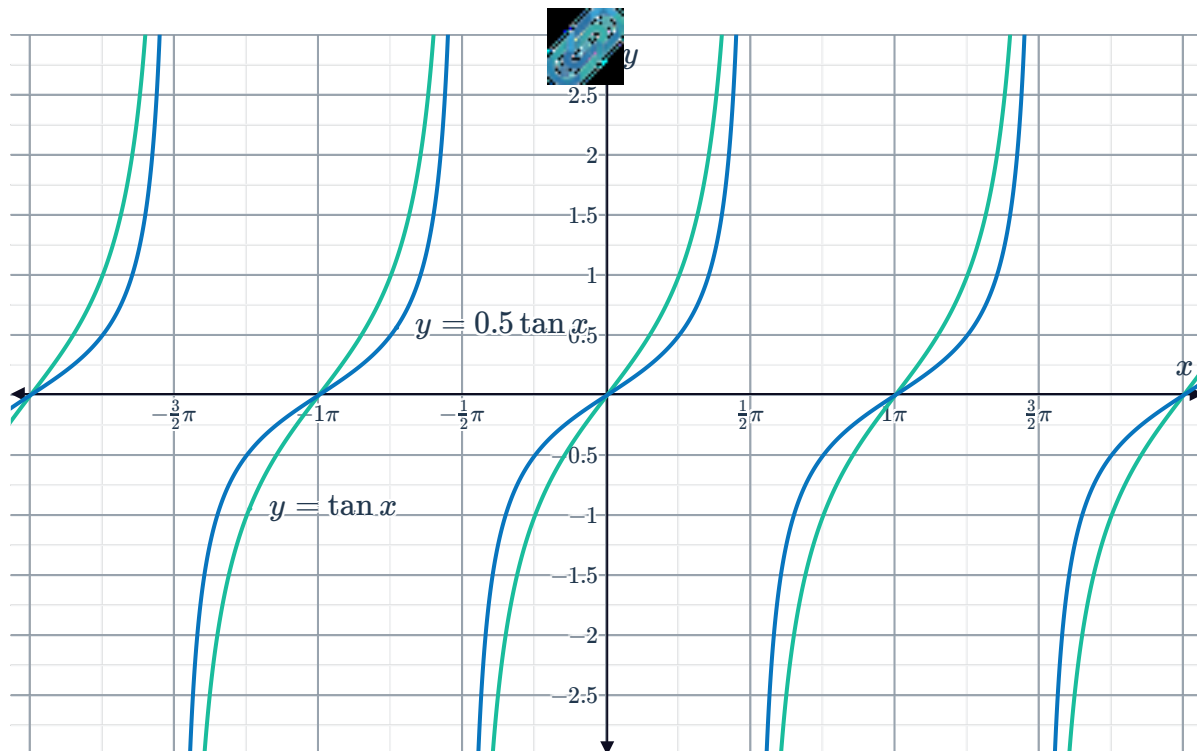
c $f(x) = 10 \tan x$

d $f(x) = -2 \tan x$

e $f(x) = -3 \tan x$

f $f(x) = 1.5 \tan x$

31



Vertical translations of the tangent function

32

- a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k .
- b $(-\infty, \infty)$

33

- a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k .
- b $(-\infty, \infty)$
- c $[3, \infty)$

34

a

i $(0, -2)$

ii -1

v $\frac{\pi}{2}$

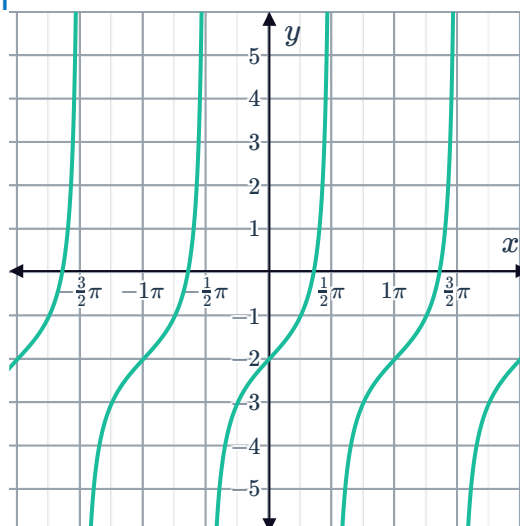
vi $-\frac{\pi}{2}$



iii π

iv π

vii



b

i $(0, 3)$

ii 8

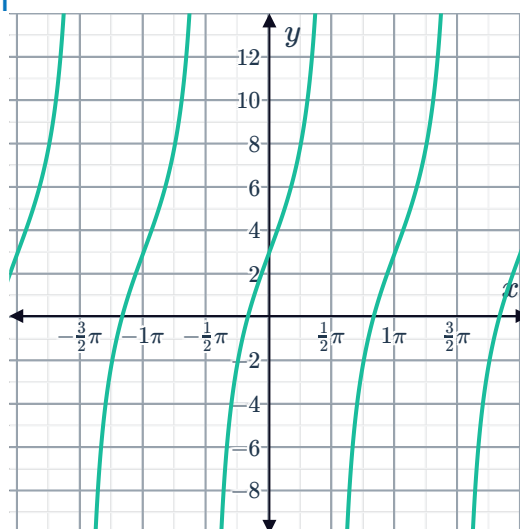
v $\frac{\pi}{2}$

vi $-\frac{\pi}{2}$

iii π

iv π

vii



35

a Translated 4 units downwards.

b $y = \tan x - 4$